

modules, and then a new feature was derived from the observed losses known as the No. of Burst between Failures – NBBF. The TCP variant of the expectation-maximisation (EM) algorithm performed better than EM with Hidden Markov Models and clustering.

Queue management is an additional mechanism in the intermediate nodes that helps with TCP congestion control mechanisms. Hence it is responsible for dropping packets whenever necessary to control the queue length in the intermediate nodes. The conventional approach for queue management was the Drop-tail mechanism. Artificial Neural Network – Active Queue Management extends the neuron Proportional-Integral-Derivative (PID) controller by including another PID. In a superficial explanation, this implementation improves the performance when compared to PID NN in real-life scenarios, thus incurring higher computational overhead.

**Resource management:** The vital resources of the network, including the CPU, memory, switches, routers and frequencies, are under resource management of a network. Here, resource allocation becomes a binary classification or decision problem, which should actively manage the resources ensuring long-term goals of resource utilisation. Admission control, which is a subdivision under resource management, was ensured by Blenk. He employed a recurrent neural network (RNN) for the online virtual network embedding (VNE) problem by predicting the probability of the virtual network request by the VNE algorithm, before running that algorithm itself. This was done based on the current state of the substrate and the request. These RNNs gained an accuracy of about 90 per cent using the previous performance data of the VNE algorithms.

## Machine learning in network security

As per the SANS Institute, network security is the process of taking preventive measures with respect to the hardware and software, to protect the underlying networking infrastructure from unauthorised access, misuse, malfunction, modification, destruction, or improper disclosure, thereby creating a secure platform for computers, users and programs to perform their permitted critical functions.

There are various specialised techniques to implement this defence. Cisco has broken down network security into the following types:

- Access control
- Anti-malware
- Application security
- Behavioural analytics
- Data loss prevention
- Email security
- Firewalls
- Intrusion detection and prevention
- Mobile device and wireless security
- Network segmentation
- Security information
- VPN
- Web security

We will now look into the solutions provided by machine learning for the prevention of various types of intrusions.

### Misuse based intrusion detection:

In misuse based detection, abnormal system behaviour is defined as making everything else normal. Hence, anything which is not known is considered normal. There have been proposals for a real-time misuse based intrusion detection system.

To reduce the number of features used, the information gain concept was used. The best technique for this purpose turned out to be the decision tree, which runs on traces collected in a 2-second interval of time and resulted in 98 per cent detection accuracy. This implementation could only detect two types of attacks -- DoS and Probe -- and still had some

vulnerabilities to persistent threats and distributed attacks. Further research has led to works that use a transductive confidence machine for a K-NN with a strangeness measure.

### Anomaly based intrusion

**detection:** The major parameter on which the anomaly detection system relies is the network behaviour. If the network behaviour is found to be within the predefined behaviour, the network transaction is accepted; otherwise, an alert gets triggered in the system.

One of the recent solutions for anomaly based intrusion detection used a support vector machine (SVM) with a radial basis function kernel. This RBF-SVM was used to devise an IDS for SDN based malware detection. Limited numbers of features like number of packets, number of bytes, flow duration, byte rate, packet rate, length of the first packet and average packet length were evaluated, and collected via SDN switches. This model resulted in 98 per cent accuracy for malware traces.

### Hybrid intrusion detection:

Apart from the above IDSs, we also have custom/hybrid IDSs, which apply both misuse based and anomaly based intrusion detection. Their sole purpose is to achieve high accuracy in detecting patterns of known attacks along with detecting new attacks in the system.

When compared to the SVM based solution, neural networks based hybrid intrusion detection systems take more training time and computational power. An SVM solution can achieve 99.5 per cent accuracy within a training time of 17.77 seconds on the KDD Cup data set, hence outperforming neural networks both with respect to accuracy and runtime. Further improvements in this field have resulted in developing a hierarchical IDS framework based on RBF for hybrid intrusion detection, reducing the complexity of the system. This implementation was also evaluated using the KDD data set against a backpropagation learning